

Visual Symbols and the Blind

Key words and phrases from the article • Level B

The words and phrases below appear in the article *Visual Symbols and the Blind*. Study the full table carefully, then complete the gap-fill exercise in Part 2. Most sentences require more than one answer.

PART 1 — VOCABULARY TABLE

WORD / PHRASE	DEFINITION	EXAMPLE SENTENCE
outlines	the shapes or outer lines of objects, used to represent them in drawings or diagrams	<i>Blind people can appreciate the use of outlines to describe the arrangement of objects in space.</i>
perspectives	techniques used in drawing to represent depth and the relationship between objects in space	<i>The use of perspectives in raised-line drawings allows blind people to understand spatial arrangements.</i>
literal representations	images or symbols that show things exactly as they appear, without metaphor or convention	<i>Pictures are more than literal representations — they also carry meaning through cultural convention.</i>
initiative	the ability to decide independently to do something without being told	<i>The blind woman decided on her own initiative to draw a wheel as it was spinning.</i>
motion lines	lines added to an illustration to suggest that something is moving	<i>Motion lines are a surprisingly recent invention in the history of illustration, first appearing around 1877.</i>
rendition	a particular version or interpretation of something; the way something is represented	<i>One particularly clever rendition appeared repeatedly: subjects showed the wheel's spokes as curved lines.</i>
spokes	the thin rods that connect the centre of a wheel to its outer rim	<i>Several blind subjects drew the wheel's spokes as curved lines to suggest it was spinning.</i>
metaphorical	using a word, image, or symbol to represent something else beyond its literal meaning	<i>When asked about their drawings, subjects described the curves as metaphorical ways of suggesting motion.</i>
idiosyncratic	peculiar or unusual to one individual; not shared by others in the same way	<i>The researcher wanted to know whether the motion lines were meaningful or merely idiosyncratic marks.</i>
perimeter	the outer edge or boundary of a shape or area	<i>Spokes extending beyond the wheel's perimeter were taken to mean the wheel had its brakes applied.</i>
raised-line drawings	tactile drawings with raised surfaces so that blind people can feel them with their fingertips	<i>The researcher created raised-line drawings of five different wheels for blind volunteers to feel.</i>

WORD / PHRASE	DEFINITION	EXAMPLE SENTENCE
volunteers	people who offer to take part in an activity or experiment without being paid or required to	<i>Eighteen blind volunteers were asked to feel the raised-line wheels and assign a motion to each one.</i>
consensus	a general agreement among a group of people about something	<i>The consensus among the sighted was barely higher than among the blind — a remarkable finding.</i>
sighted	having the ability to see; not blind	<i>The control group consisted of eighteen sighted undergraduates from the University of Toronto.</i>
doctoral	relating to a doctorate — the highest level of university degree	<i>With Chang Hong Liu, a doctoral student from China, the author began exploring symbolic shapes.</i>
symbolism	the use of symbols to represent ideas or qualities; the meaning conveyed through symbols	<i>They began exploring how well blind people understand the symbolism behind shapes like hearts.</i>
ascribed	attributed or assigned a particular quality or meaning to something	<i>A full 94% of subjects ascribed happy to the circle rather than to the square.</i>
abstract shapes	geometric forms such as circles and squares, considered apart from any specific object	<i>The researchers concluded that blind people interpret abstract shapes in the same way as sighted people.</i>
visual metaphors	images or symbols that represent something other than what they literally show	<i>The researchers found that blind people understand visual metaphors as well as sighted people do.</i>
transcends	goes beyond the usual limits or boundaries of something	<i>The symbolic meaning of abstract shapes appears to transcend visual experience entirely.</i>
arbitrary	based on random choice or personal whim rather than any clear principle or reason	<i>If blind people understand motion lines instinctively, then those conventions are not arbitrary at all.</i>
implications	possible consequences or results that are not directly stated but can be inferred	<i>These findings carry important practical implications for the design of materials for blind people.</i>

PART 2 — GAP-FILL EXERCISE

Complete the sentences using the correct word or phrase from the word bank below. You may need to change the form of the word. Most sentences require more than one answer.

WORD BANK

outlines	perspectives	literal representations	initiative
motion lines	rendition	spokes	metaphorical
idiosyncratic	perimeter	raised-line drawings	volunteers
consensus	sighted	doctoral	symbolism
ascribed	abstract shapes	visual metaphors	transcends
arbitrary	implications		

1	Blind people can appreciate the use of _____ and _____ to describe the arrangement of objects — but pictures are more than _____, and this is what makes the research so interesting.
2	On her own _____, a blind woman drew a spinning wheel using _____ — a device that, as art scholar David Kunzle noted, was barely used in illustration before 1877.
3	One particularly clever _____ appeared repeatedly: several subjects drew the wheel's _____ as curved lines, describing them as _____ ways of suggesting movement.
4	The researcher wanted to know whether the curves were genuinely meaningful or merely _____ — and whether blind and _____ subjects would interpret the lines in the same way.
5	He created _____ of five different wheels and asked eighteen blind _____ to feel them, while a control group of _____ undergraduates did the same task visually.
6	Spokes extending beyond the wheel's _____ were taken to mean the brakes were applied; dashed spokes suggested the wheel was spinning quickly — interpretations shared by both groups.
7	The _____ among the sighted was barely higher than among the blind, showing that the blind not only interpreted _____ but did so with nearly identical accuracy.
8	With a _____ student from China, the author began investigating the _____ behind shapes such as hearts, which do not directly represent their meaning.
9	A full 94% of subjects _____ happy to the circle rather than the square — suggesting that our associations between _____ and concepts like happiness are widely shared.
10	When blind volunteers completed the same task, their responses closely resembled the _____ reached by _____ subjects, confirming that they interpret _____ in the same way.
11	If blind people understand _____ as intuitively as sighted people, then the conventions of illustration cannot be _____ — they must reflect deeper patterns in how the mind works.
12	These findings carry important _____ for the design of materials for blind people, suggesting that tactile images may be far more intuitive than previously assumed — because the symbolic language of pictures _____ visual experience entirely.

Write your answers here:

1. _____	2. _____	3. _____
4. _____	5. _____	6. _____
7. _____	8. _____	9. _____
10. _____	11. _____	12. _____